

Realiza las siguientes tareas:

- En tu equipo local, **crea un contenedor** con la imagen de ubuntu 22.04. Instala el servicio web y crea una web con tu nombre y apellidos en una etiqueta **h1**.

```
alumnado@iespsur: ~  
alumnado@iespsur: ~ 102x32  
alumnado@iespsur:~$ docker run -it --name WebNueva -p 8080:80 ubuntu:24.04 /bin/bash  
root@8309070b6314:/# apt update  
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]  
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]  
Get:6 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]  
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1138 kB]  
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.6 kB]  
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2462 kB]  
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1512 kB]  
Get:12 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]  
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2605 kB]  
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1879 kB]  
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [38.9 kB]  
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1924 kB]  
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [33.9 kB]  
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.4 kB]  
Fetched 33.9 MB in 8s (4241 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
1 package can be upgraded. Run 'apt list --upgradable' to see it.  
root@8309070b6314:/# apt install apache2  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  adduser apache2-bin apache2-data apache2-utils ca-certificates krb5-locales libapr1t64  
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 libbrotli1 libcurl4t64 libexpat1  
root@8309070b6314:/# service apache start  
apache: unrecognized service  
root@8309070b6314:/# service apache2 start  
* Starting Apache httpd web server apache2  
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.  
0.2. Set the 'ServerName' directive globally to suppress this message  
*  
root@8309070b6314:/# echo '<!doctype html><html><head><meta charset="utf-8"><title>Mi web</title></hea  
d><body><h1>Jesus Parra</h1></body></html>' > /var/www/html/index.html  
root@8309070b6314:/# chown www-data:www-data /var/www/html/index.html  
root@8309070b6314:/# chmod 644 /var/www/html/index.html  
root@8309070b6314:/# exit  
alumnado@iespsur:~$ docker login
```

- Crea una imagen del contenedor en tu equipo local.

```
alumnado@iespsur:~$ docker commit WebNueva imagenjesus:web  
sha256:fede9f01c1563189474e3d4980c1f7b2c84d380fde171ff3a4070005425cbff1  
alumnado@iespsur:~$ docker images  
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE  
imagenjesus         web            fede9f01c156    7 seconds ago  246MB  
ubuntu              24.04         ce8f79aecc43    2 days ago     78.1MB  
alumnado@iespsur:~$ docker push charcutron/imagenjesus:web  
The push refers to repository [docker.io/charcutron/imagenjesus]  
An image does not exist locally with the tag: charcutron/imagenjesus  
alumnado@iespsur:~$ docker tag imagenjesus:web charcutron/imagenjesus:web  
alumnado@iespsur:~$ docker push charcutron/imagenjesus:web  
The push refers to repository [docker.io/charcutron/imagenjesus]  
ac837a7f52dd: Preparing  
65b08cd99c60: Preparing  
denied: requested access to the resource is denied  
alumnado@iespsur:~$
```

Me salio un error de logueo por lo que tuve que verificar el mail y volver a intentarlo

```
alumnado@iespsur:~$ docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: BGGW-LSVL
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/alumnado/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
alumnado@iespsur:~$ docker commit WebNueva imagenjesus:web
sha256:15bc3925ac20e13656676f3b087c416c4bb56c5c7ddae22dd17736cd1b85a5
alumnado@iespsur:~$ docker tag imagenjesus:web charcutron/imagenjesus:web
alumnado@iespsur:~$ docker push charcutron/imagenjesus:web
The push refers to repository [docker.io/charcutron/imagenjesus]
ac837a7f52dd: Pushed
65b08cd99c60: Mounted from library/ubuntu
web: digest: sha256:c887a4f384a4ae9a571af8a8582dee9fcb9be9602fbdf4e306e0475e9c5bb520 size: 741
alumnado@iespsur:~$
```

- Borra el contenedor anterior y crea uno nuevo **partiendo de la imagen que has creado anteriormente**. Comprueba que la web es visible en un navegador de tu equipo local.

```
alumnado@iespsur: ~
alumnado@iespsur: ~ 102x32
alumnado@iespsur:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
8309070b6314   ubuntu:24.04   "/bin/bash"             14 minutes ago Exited (0)    11 minutes ago WebNueva
alumnado@iespsur:~$ docker rm WebNueva
WebNueva
alumnado@iespsur:~$ docker run -d --name WebNueva2 -p 8080:80 charcutron/imagenjesus:web
86b27800ad86f323610501b17d4e2f1a8ea190c6d4c71d2046e25f5f786ea002
alumnado@iespsur:~$
```

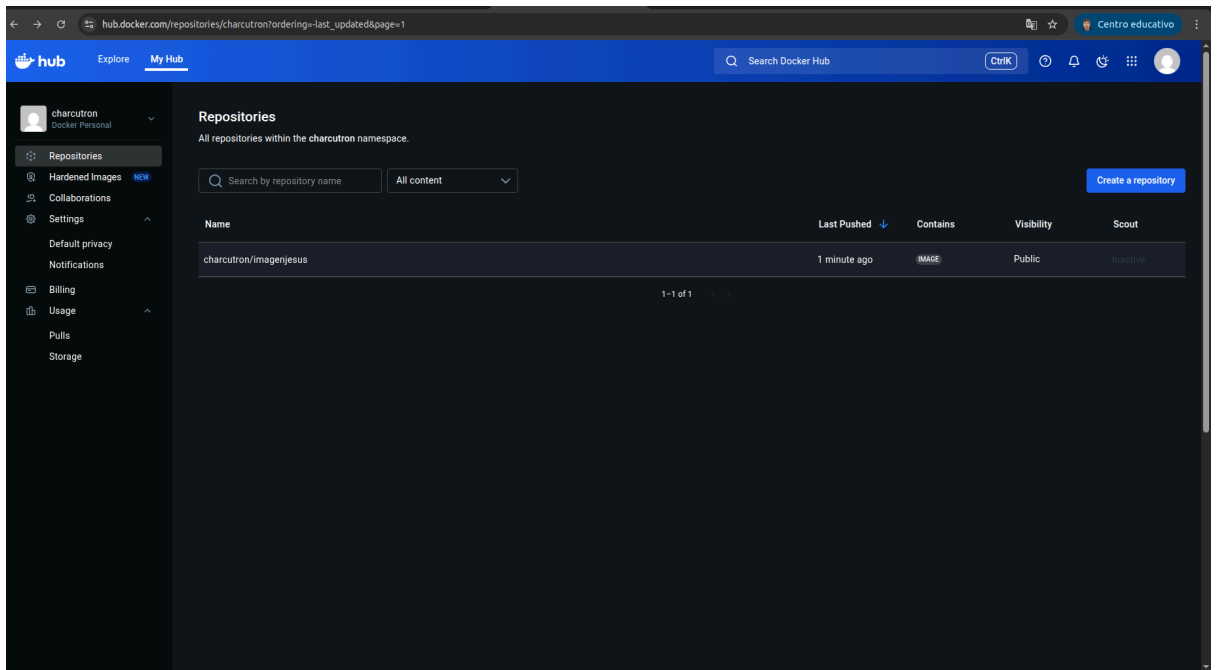
Habia creado la imagen pero habia usado -d y eso me estaba dando errores hasta que me he dado cuenta y lo he cambiado por -it

```
root@ada35821d993: /
root@ada35821d993: / 102x32
PORTS      NAMES
86b27800ad86 charcutron/imagenjesus:web  "/bin/bash"  3 minutes ago  Exited (0) 4 seconds ago
WebNueva2
alumnado@iespsur:~$ docker rm WebNueva2
WebNueva2
alumnado@iespsur:~$ docker run -d --name WebNueva2 -p 80:80 charcutron/imagenjesus:web /bin/bash
8432c60bb0347be44de2dfd8443e9f8973ac0f55c3917833c3477ace789f7eab
alumnado@iespsur:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS         NAMES
8432c60bb034   charcutron/imagenjesus:web         "/bin/bash"            6 seconds ago  Exited (0) 5 seconds ago
WebNueva2
alumnado@iespsur:~$ docker start -a WebNueva2
alumnado@iespsur:~$ docker start WebNueva
Error response from daemon: No such container: WebNueva
Error: failed to start containers: WebNueva
alumnado@iespsur:~$ docker start WebNueva2
WebNueva2
alumnado@iespsur:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS         NAMES
8432c60bb034   charcutron/imagenjesus:web         "/bin/bash"            41 seconds ago  Exited (0) 4 seconds ago
WebNueva2
alumnado@iespsur:~$ docker rm WebNueva2
WebNueva2
alumnado@iespsur:~$ docker run -it --name WebNueva2 -p 80:80 charcutron/imagenjesus:web /bin/bash
root@ada35821d993:/# service apache2 start
* Starting Apache httpd web server apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress this message
*
root@ada35821d993:/#
```

← → 🔍 localhost ☆ Centro educativo

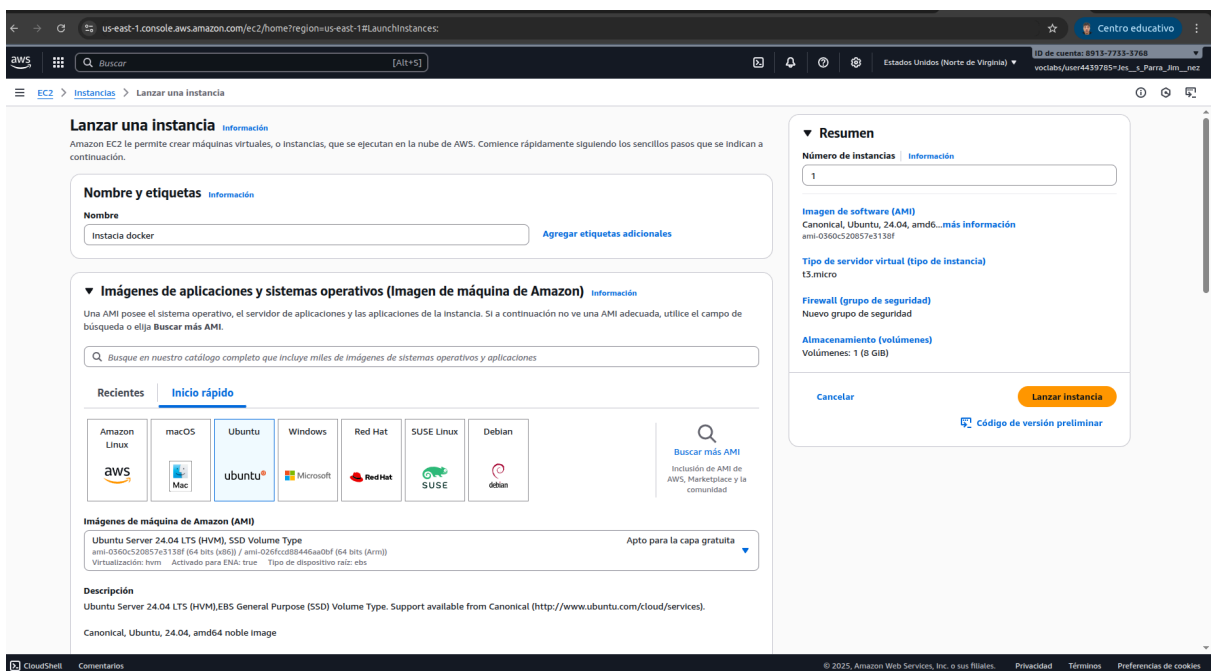
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- Si todo está bien, sube la imagen que has creado a **Docker Hub**.



- Crea una nueva nueva **instancia EC2 AWS** (Ubuntu 22.04, 1GB Ram, 8 GB HD)

- **Nos metemos en el laboratorio de AWS y creamos la instancia:**



```
alumnado@iespsur: ~/Despliegue Web
alumnado@iespsur:~/Despliegue Web 102x32
alumnado@iespsur:~$ cd descargas
bash: cd: descargas: No existe el archivo o el directorio
alumnado@iespsur:~$ cd Descargas
alumnado@iespsur:~/Descargas$ ssh -i "daweb2025.pem" ubuntu@ec2-54-163-71-86.compute-1.amazonaws.com
Warning: Identity file daweb2025.pem not accessible: No such file or directory.
The authenticity of host 'ec2-54-163-71-86.compute-1.amazonaws.com (54.163.71.86)' can't be established.
ED25519 key fingerprint is SHA256:9KQ4/BerhZiGIy7kV1HqCtFwm0sBk2XfPCy7TePVm9o.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-163-71-86.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
ubuntu@ec2-54-163-71-86.compute-1.amazonaws.com: Permission denied (publickey).
alumnado@iespsur:~/Descargas$ ssh -i "daweb2025.pem" ubuntu@ec2-54-163-71-86.compute-1.amazonaws.com
Warning: Identity file daweb2025.pem not accessible: No such file or directory.
ubuntu@ec2-54-163-71-86.compute-1.amazonaws.com: Permission denied (publickey).
alumnado@iespsur:~/Descargas$ ec2-54-163-71-86.compute-1.amazonaws.com
ec2-54-163-71-86.compute-1.amazonaws.com: orden no encontrada
alumnado@iespsur:~/Descargas$ cd
alumnado@iespsur:~$ cd Despliegue Web
bash: cd: demasiados argumentos
alumnado@iespsur:~$ cd Despliegue\ Web/
alumnado@iespsur:~/Despliegue Web$ ec2-54-163-71-86.compute-1.amazonaws.com
ec2-54-163-71-86.compute-1.amazonaws.com: orden no encontrada
alumnado@iespsur:~/Despliegue Web$ ssh -i "daweb2025.pem" ubuntu@ec2-54-163-71-86.compute-1.amazonaws.com
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

- **Despliega el contenedor** en la nueva instancia EC2 (debes usar el usuario **ubuntu** para gestionar el servicio Docker).

El comando que he utilizado para crear el contenedor dentro de la instancia ha sido:

`docker run -d --name paginaWeb -p 80:80 charcutron/imagenjesus:web apache2ctl -D FOREGROUND`, este apartado que significa: `apache2ctl -D FOREGROUND`

El cual lanza el contenedor en segundo plano, asignándole un nombre y exponiendo el puerto 80. Apache se inicia y permanece en primer plano dentro del contenedor gracias a `apache2ctl -D FOREGROUND`, garantizando que el contenedor siga ejecutándose y que la web esté accesible de forma remota.

```
ubuntu@ip-172-31-25-12: ~
ubuntu@ip-172-31-25-12:~ 168x31
ubuntu@ip-172-31-25-12:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
4df8b64fff5    charcutron/imagenjesus:web          "apache2ctl -D FOREG..." 5 minutes ago  Up 5 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp  paginaWeb
ubuntu@ip-172-31-25-12:~$
```

- **Comprueba desde tu equipo local** que puedes ver la web remota con tu nombre.



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```
ubuntu@ip-172-31-25-12:~$ curl http://54.87.224.242/  
<!doctype html><html><head><meta charset="utf-8"><title>Mi web</title></head><body><h1>Jesus Parra</h1></body></html>  
ubuntu@ip-172-31-25-12:~$
```

Puedes encontrarte alguno de estos problemas frecuentes o cualquier otro. **Es importante que documentes lo que te ha ocurrido y cuál es la solución que has encontrado:**

- **403/timeout** al abrir la web
- **“permission denied”** usando docker con ubuntu
- No existe la imagen al hacer **pull**
- Página por defecto de Apache en lugar de tu **index**